a From source assertions to a historical question

Modern paternal evidence

Reported origins → reconciled units
Branch paths and multiple localities
Sampling and dependence retained

Dated archaeological evidence

Context, material, date and geometry
Original assertions + source corrections
Period and feature support checked

A common measurement and explicit contrasts

Nearest-feature proximity → fixed archaeological basis
Unit-level locality mixture → branch and direct period contrasts
Separate profiles retain their separate archaeological meanings

Dated people and activity context

Ancient genomes supply dated people
Activities checked against reports
Dates, low coverage and kinship visible

Historical appraisal

Formation, redistribution or later sorting?
Assess agreement, alternatives and gaps
Modern concordance does not date descent

b Mannaberg: one context enters a landscape measurement

Reported cultivation context
A1462 · ash charcoal · Ua-55701
3449 ± 30 BP; 1880–1680 BCE
Crop remains are not directly dated

Closest represented nominal origin: 6.27 km
Site proximity $\exp(-d / 50) = 0.882$
All-site mean clearance proximity = 0.786
Full seven-family Component 3 = 1.425

The numerical example uses all eligible sites and paired date/location draws for the full score.
A dated ash sample qualifies the context; it does not identify the occupants' paternal lineage.